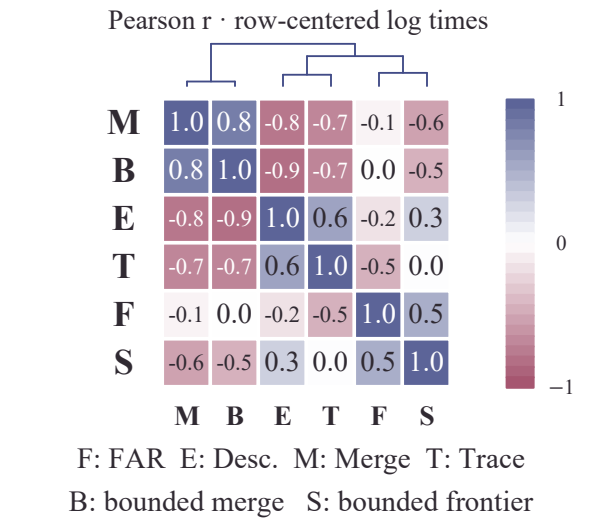


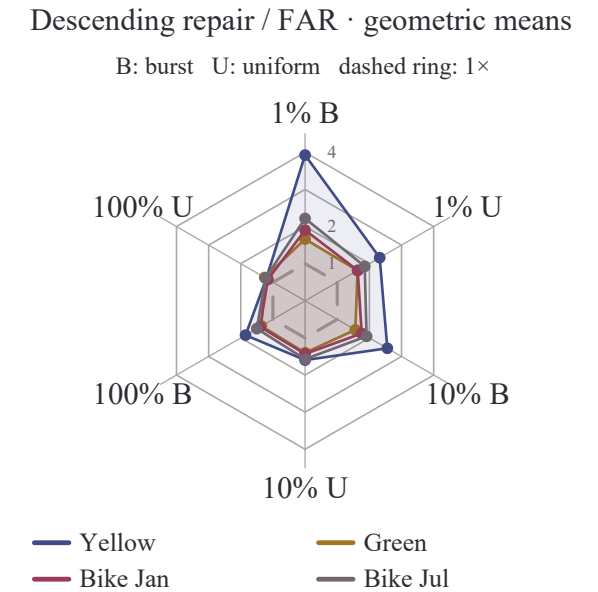
Clustered runtime profiles



One equality-key group

$q = [8, 12, 15, 19, 24, 28]$
 $y = [4, 10, 10, 18, 18, 27]$
 $D = [11, 13, 14, 23, 26]$

Workload performance radar

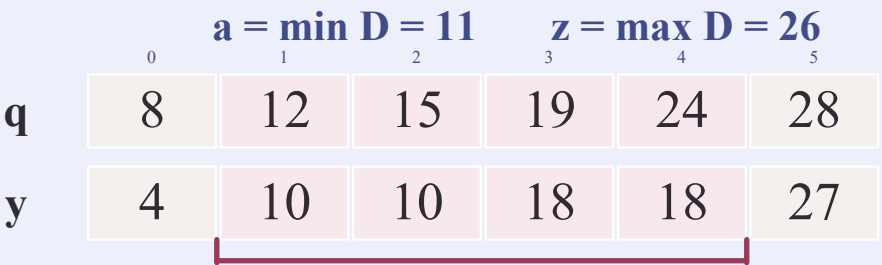


Inputs and measured evidence

Fixed probes + late insertions: exact ASOF maintenance

FAR: use ordered answers to bound and route repair

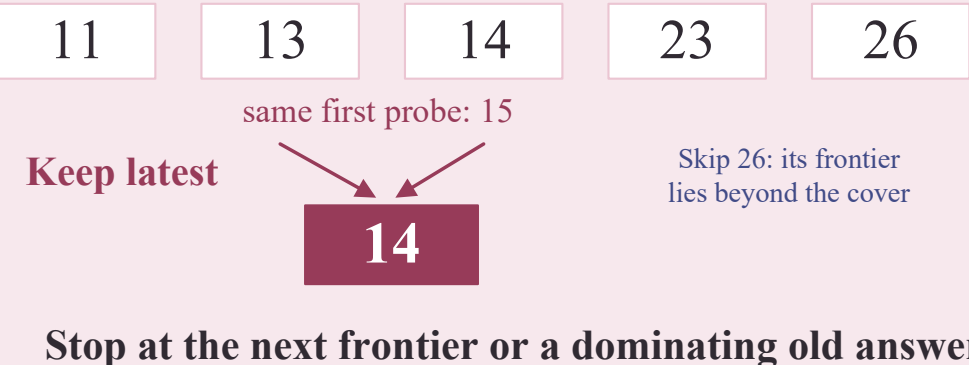
1 Bound repair using the cached answers



$L = \text{lb}(q, a) = 1$ $H = \text{lb}(y, z) = 5$

Only $[L, H)$ can change; ranks are zero-based

Sparse branch: compact events at probe frontiers



2 Route by candidate span: $s = H - L, b = |D|$



Answer bounds → probe frontiers → adaptive traversal

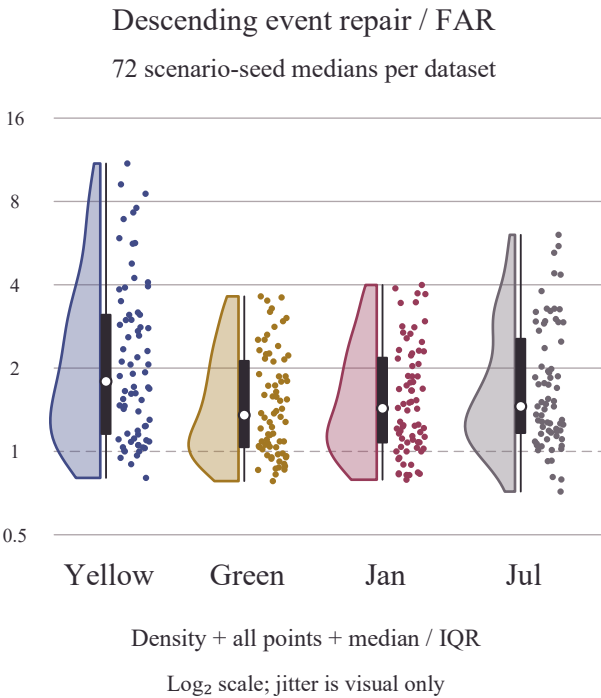
Exact maintained output

Worked example · 3 changed answers

Probe	Old	New
8	4	4
12	10	11
15	10	14
19	18	18
24	18	23
28	27	27

Illustrative timestamps, not measurements

Speedup rainclouds



Outputs and evaluation